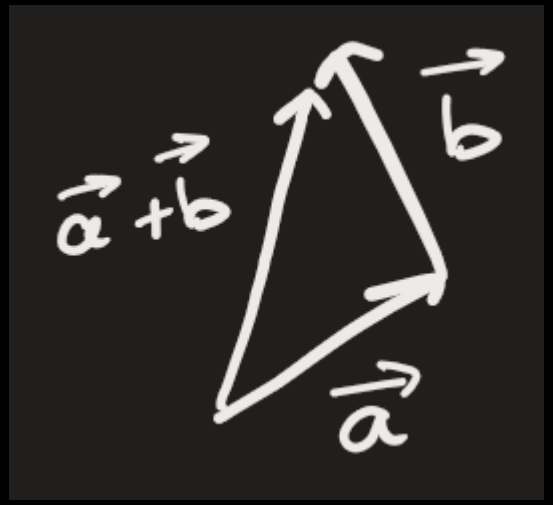
Vector addition:



$$\vec{a} = (2, 3, 6)$$
$$\vec{b} = (3, 2, 5)$$

Here $\vec{a}$ + $\vec{b}$ = (2+3, 3+2, 6+5) = (5, 5, 11)

Example:

1) basket1 + basket2 = basket_total

(2 apple, 4 banana, 1 Kiwi) + (3 apple, 2 banana, 4 Kiwi) = (5 apple, 6 banana, 5 Kiwi)

$\Rightarrow$ (2,4,1) + (3,2,4) = (5, 6, 5)

Condition: The size of vectors should be same

Example:

$$\vec{a} = (2, 3, 6)$$
$$\vec{b} = (3, 2, 5, 7, 8)$$

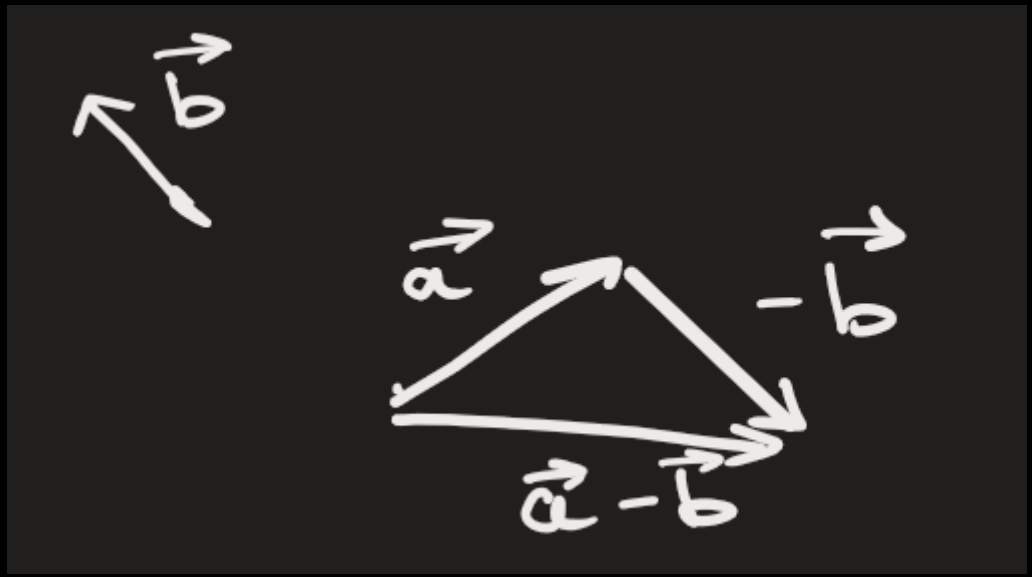
Here $\vec{a}$ + $\vec{b}$ = NOT possible

Vector subtraction:

$$\vec{a} = (2, 3, 6)$$
$$\vec{b} = (3, 2, 5)$$

Here $\vec{a}$ - $\vec{b}$ = (2 - 3, 3 - 2, 6 - 5) = (-1, 1, 1)

Condition: The size of vectors should be same



Scalar multiplication:

Scalar x vector:

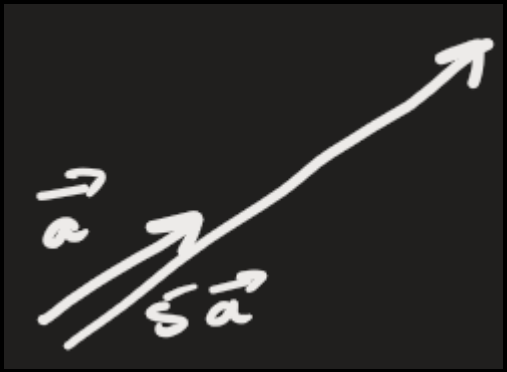
If $\vec{a}$ = (2, 4, 7), then

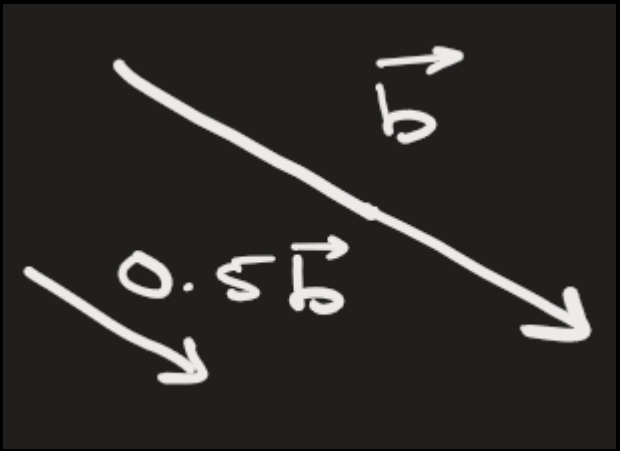
$\mathbf{5}\vec{a}$

= 5 x (2, 4, 7)

= (5x2, 5x4, 5x7)

= (10, 20, 35)





Example: **basket** = (2 apple, 4 banana)

5 x **basket**

= 5 x (2 apple, 4 banana)

= 5 x (2, 4)

= (10, 20)

= (10 apple, 20 banana)

Dot product of vectors:

For 2 vector **a** and **b:**

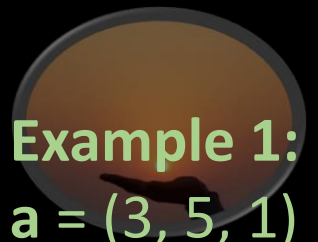
$$\mathbf{a} = [a_1, a_2, a_3]$$

$$\mathbf{b} = [b_1, b_2, b_3]$$

The dot product is

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

Condition for dot product: The size of 2 vectors should be same



Example 1:

a = (3, 5, 1)

b = (2, 4, -7)

Then,

a.b

= (3, 5, 1) . (2, 4, -7)

= 3x2 + 5x4 + 1x-7

= 6 + 20 -7 = 19

Example2:

a = (3, 5, 1)

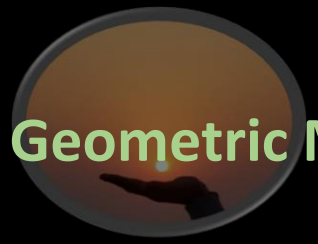
b = (2, 4, -7, 6)

Dot product not possible.

$$\mathbf{a} = [a_1, a_2, a_3]$$

$$\mathbf{b} = [b_1, b_2, b_3]$$

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$



Geometric Meaning (Deep Intuition)

The dot product measures **how aligned two vectors are**

$$\mathbf{a} = [a_1, a_2, a_3]$$

$$\mathbf{b} = [b_1, b_2, b_3]$$

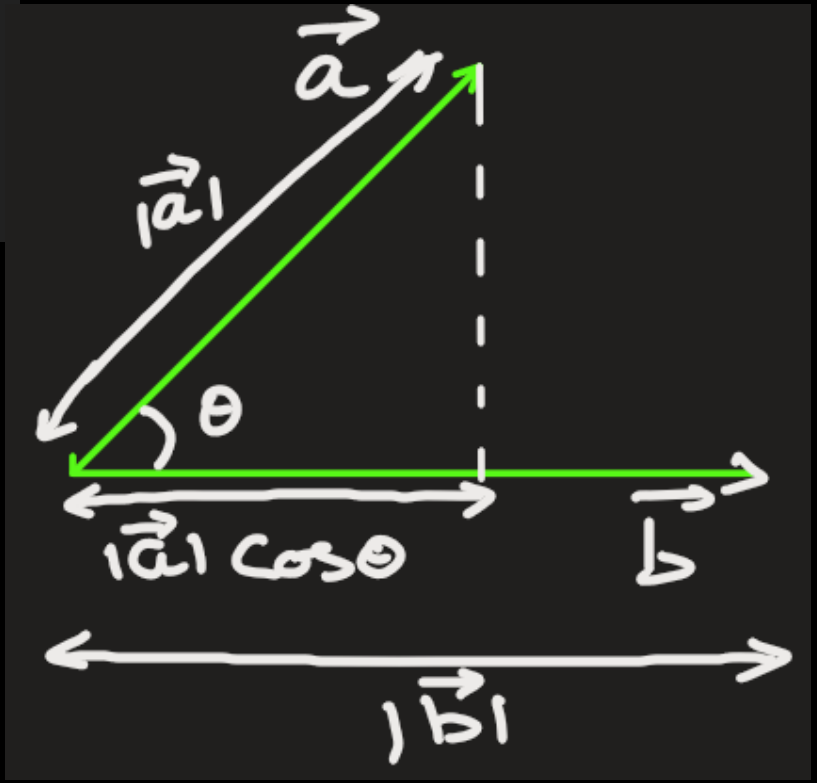
$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

Mathematically:

$$a \cdot b = |a||b| \cos(\theta)$$

Where:

- θ = angle between vectors

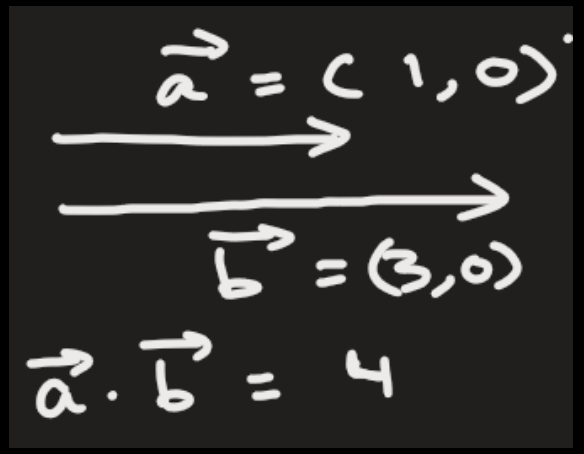


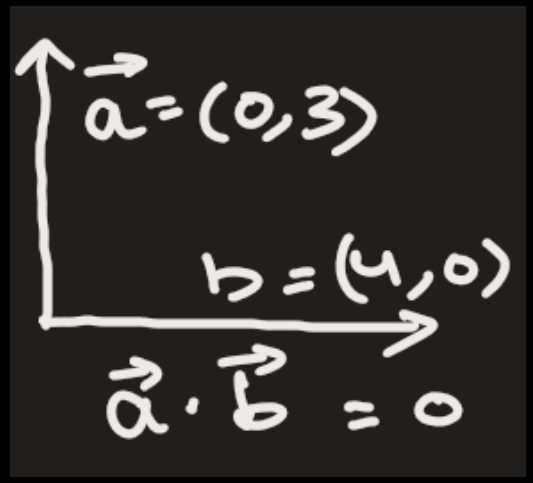
Angle Between Vectors	Dot Product Meaning
0°	Very similar direction (large positive)
90°	Unrelated (0)
180°	Opposite meaning (negative)

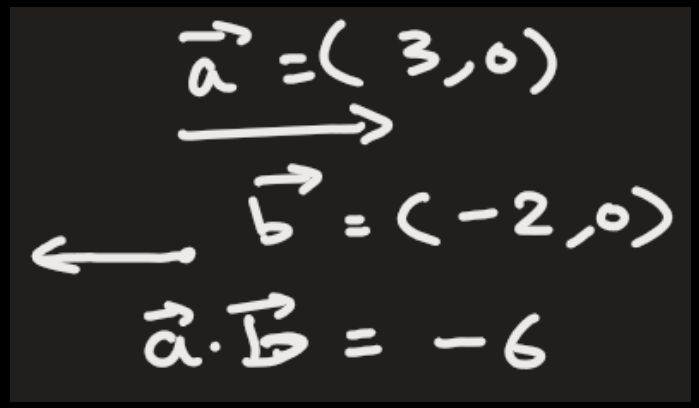
$$\mathbf{a} = [a_1, a_2, a_3]$$

$$\mathbf{b} = [b_1, b_2, b_3]$$

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$







Dot Product in Machine Learning (Core Idea)

In ML:

One vector = **data (features)**

Another vector = **model parameters (weights)**

Dot product = **model decision score**

For example:

x = [size of rooms, number of bedrooms, age of house]

w = [size_weight, bedroom_weight, age_weight]

Prediction on price = w·x

$$\vec{x} = (200 \text{ sq.ft}, 6 \text{ bedrooms}, 10 \text{ year})$$

$$\vec{w} = (0.1, 5, 2.5)$$

$$y = \vec{w} \cdot \vec{x}$$

$$= 200 \times 0.1 + 6 \times 5 + 10 \times 2.5$$

$$= 75$$

Dot Product to find similarity between 2 vectors

Cosine Similarity: $\cos\theta = \dfrac{\vec{a} \cdot \vec{b}}{|\vec{a}|\,|\vec{b}|}$

doc_a = "Today is beautiful day" $\rightarrow \vec{a} = (0.1, 0.3, \ldots)$

doc_b = "Today is my lucky day" $\rightarrow \vec{b} = (0.5, 0.2, \ldots)$

doc_c = "The food made me sick" $\rightarrow \vec{c} = (0.7, 0.8, \ldots)$

$$\cos\theta_{ab} = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|} = 0.7 \text{ (suppose)}$$

$$\cos\theta_{ac} = \frac{\vec{a}\cdot\vec{c}}{|\vec{a}||\vec{c}|} = 0.2 \text{ (suppose)}$$

$$\cos\theta_{bc} = \frac{\vec{b}\cdot\vec{c}}{|\vec{b}||\vec{c}|} = 0.3 \text{ (suppose)}$$

Conclusion:

- doc_a and doc_b are similar.
- doc_a and doc_c are not similar